

EX. NO: 3

Simple AI Agent (Rule-based)

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Aim:

To create a simple rule-based AI agent in Python that responds to user inputs with predefined replies based on keyword matching

Procedure:

1. Design the AI logic:

Use rule-based (reflex) logic where specific keywords or phrases in user input trigger predefined responses.

2. Normalize user input:

Convert the user's input to lowercase to simplify matching rules regardless of capitalization.

3. Define response rules:

Create conditional checks (if-elif-else) for key phrases such as greetings, asking the agent's name, weather queries, and exit commands.

4. Implement a main loop:

Continuously prompt the user for input, pass it to the AI agent function, and print the response.

5. Exit mechanism:

Allow the user to type 'bye' to end the conversation and exit the program.

```
def simple_ai_agent(user_input):  
  
    # Convert input to lowercase for easy matching  
    user_input = user_input.lower()  
  
    # Define rules (reflex-based logic)  
  
    if "hello" in user_input or "hi" in user_input:  
  
        return "Hello! How can I help you today?"
```

```

elif "your name" in user_input:

    return "I'm a simple AI agent created in Python."

elif "weather" in user_input:

    return "I can't check real-time weather, but I hope it's nice where you are!"

elif "bye" in user_input:

    return "Goodbye! Have a great day!"

else:

    return "I'm not sure how to respond to that."

# Main loop

print("Welcome to the Simple AI Agent. Type 'bye' to exit.")

while True:

    user_input = input("You: ")

    response = simple_ai_agent(user_input)

    print("AI: " + response)

    if "bye" in user_input.lower():

        break

```

Output:

```

Welcome to the Simple AI Agent. Type 'bye' to exit.
You: Hi
AI: Hello! How can I help you today?
You: What's your name?
AI: I'm a simple AI agent created in Python.
You: How's the weather?
AI: I can't check real-time weather, but I hope it's nice where you are!
You: Tell me a joke.
AI: I'm not sure how to respond to that.
You: bye

```

AI: Goodbye! Have a great day!

Result:

Thus, the simple AI agent was successfully implemented using a rule-based approach.